



Ahmedabad University
School of Engineering and Applied Science

Brain Tumor Segmentation and Localization Using Traditional Image Processing and OpenCV

Weekly Status Report Week 1 of 10

Course:	ECE501 – Digital Image Processing	Group No.:	18
Instructor:	Professor Mehul Raval	Project No.:	20
Section:	01	Report Date:	12/09/2026

Repository: github.com/codesensei404/DIP_2026_Project20_Group18

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Keywords: Brain Tumor, Image Segmentation, OpenCV, Image Processing, Thresholding.

1 Introduction

For this project, we will be trying to first localize the Brain Tumor from MRIs (Magnetic Resonance Imaging) using Classic Image Processing Techniques. We will be only using Python language and some open source Python libraries and frameworks such as OpenCV, NumPy, Matplotlib etc.

2 Objective

This is the first week and the kick starter week of our project, in this week our main objective was to define the problem statement and start working on designing the structural framework of our whole Project. This is a very well known research area and there are a lot of resources online for this field.

It does make it easy for us to utilize the online resources and IEEE Databases for research papers, while on the other hand we are restricted with only using Traditional Image Processing Techniques such as CLAHE (Contrast Limited Adaptive Histogram Equalization) for amplifying noise in flat areas, Watershed Segmentation for boundary and perimeter detection etc. For the process definition of our project we have chosen our first 3 papers as starters which are closely connected with classical image processing techniques and getting very close results. The main ideas of these papers and their methodologies are discussed in the next section.

3 Literature Review

In the first paper, a hybridized approach is used where multilevel thresholding is used for the first part for the segmentation, which completely eliminates the need for manual seeding and it results in the potential tumor regions getting recognized. To better refine the shape and the boundary of the tumor a level set method with some fine tuned morphological operations are applied, which results in a more precise shape. [1]

While the second paper closely resembles the methods we learned in the class in week 2 and 3 about histogram equalization. In the second paper the data is preprocessed via histogram equalization for increasing the contrast for better segmentation. Then for the segmentation part the authors have used Global Thresholding with morphological functions such as Erosion and Dilation to further isolate the tumor region from healthy tissues of brain. Lastly to get the fine tuned quantitative parameters a rigorous mathematical computation pipeline is constructed which successfully results in finding the area, parameter and the centroid of the tumor region. [2]

The third paper we chose was more on the tumor severity detection. The main objective of the authors in this paper was to establish a standard pipeline for a complete CAD (Computer Aided Diagnosis) system. At the base of which they used Histogram Equalization for the initial phase and adaptive thresholding for segmentation with morphological operations. Their final testing was done on BRATS (Multimodal Brain Tumor Image Segmentation) Dataset. They successfully achieved a 86.84% of detection accuracy rate on the BRATS benchmark while providing detailed spatial and geometric output. [3]

4 Work Completed This Week

In this week, we successfully defined the project boundaries. We found 3 research papers to get idea of what methods can be used and what could be the different phases of execution including but not limited to preprocessing, segmentation and morphological operations as well as final mathematical pipeline for Tumor Region parameters.

5 Challenges Faced

Finding the Research Papers closely aligned with our project definitions and classical image processing methods.

Choosing the correct dataset for our project pipeline and we still have not found a BRATS Dataset for that but we are currently working on that.

Defining the clear project scope for removing future conflicts.

6 Plan for Next Week

In the next week our goal is to define and plan different phases of our multiphase pipeline. Where the first phase is about preprocessing the data, the second phase is the crucial segmentation part for localization and the third phase is the Tumor Parameters computation using mathematical analysis.

We will also be trying to find a suitable BRATS Dataset which is closely related to our methodology and will add that in the github repository.

References

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